

```
import pandas as pd
import numpy as np
```

```
data = pd.read_csv("/content/fraudTest.csv")
```

(555719, 23)

```
Index(['Unnamed: 0', 'trans_date_trans_time', 'cc_num', 'merchant', 'category',
      'amt', 'first', 'last', 'gender', 'street', 'city', 'state', 'zip',
      'lat', 'long', 'city_pop', 'job', 'dob', 'trans_num', 'unix_time',
      'merch_lat', 'merch_long', 'is_fraud'],
      dtype='object')
```

```
data.columns = data.columns.str.strip()
```

[illegible]

```
print(data.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 555719 entries, 0 to 555718
Data columns (total 23 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0             555719 non-null  int64
1   trans_date_trans_time  555719 non-null  object
2   cc_num                 555719 non-null  int64
3   merchant              555719 non-null  object
4   category              555719 non-null  object
5   amt                   555719 non-null  float64
6   first                 555719 non-null  object
7   last                  555719 non-null  object
8   gender                555719 non-null  object
9   street                555719 non-null  object
10  city                  555719 non-null  object
11  state                 555719 non-null  object
12  zip                   555719 non-null  int64
13  lat                   555719 non-null  float64
```

```

14 long          555719 non-null float64
15 city_pop      555719 non-null int64
16 job           555719 non-null object
17 dob           555719 non-null object
18 trans_num     555719 non-null object
19 unix_time     555719 non-null int64
20 merch_lat     555719 non-null float64
21 merch_long    555719 non-null float64
22 is_fraud      555719 non-null int64
dtypes: float64(5), int64(6), object(12)
memory usage: 97.5+ MB
None

```

```
print(data.isnull().sum())
```

```

Unnamed: 0      0
trans_date_trans_time  0
cc_num          0
merchant        0
category        0
amt             0
first           0
last            0
gender          0
street          0
city            0
state           0
zip             0
lat             0
long            0
city_pop        0
job             0
dob             0
trans_num       0
unix_time       0
merch_lat       0
merch_long      0
is_fraud        0
dtype: int64

```

Fraud and Genuine Transaction Count

```

fraud_count = data['is_fraud'].value_counts()

print("\nTransaction Count:")
print(fraud_count)

print("\nExplanation:")
print("0 = Genuine Transaction")
print("1 = Fraudulent Transaction")

```

```

Transaction Count:
is_fraud
0    553574
1     2145
Name: count, dtype: int64

Explanation:
0 = Genuine Transaction
1 = Fraudulent Transaction

```

Features and Targets

```

X= data.drop(columns=[
    'Unnamed: 0', 'is_fraud', 'merchant', 'category', 'first', 'last',
    'gender', 'street', 'city', 'state', 'job', 'dob', 'trans_num'
], axis=1)
y = data['is_fraud']

```

Splitting the dataset

```
from sklearn.model_selection import train_test_split
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

```

```
X_train['trans_date_trans_time'] = pd.to_datetime(X_train['trans_date_trans_time']).astype(int) / 10**9
```

Creating the Model

```
from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression(max_iter=5000)
```

Training the Model

```
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=5000)
```

```
X_test['trans_date_trans_time'] = pd.to_datetime(X_test['trans_date_trans_time']).astype(int) / 10**9
```

Predicting Transactions

```
y_pred = model.predict(X_test)
```

Checking the Accuracy

```
from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:")
print(accuracy)
```

```
Accuracy:
0.9961671345281796
```

```
from sklearn.metrics import confusion_matrix
```

```
print("\nClassification Report:")
print(classification_report(y_test, y_pred, zero_division=0))

cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
```

```
Classification Report:
              precision    recall  f1-score   support

     0       1.00        1.00        1.00    110718
     1       0.00        0.00        0.00        426

 accuracy          0.99
 macro avg         0.50
 weighted avg      0.99
```

```
Confusion Matrix:
[[110718    0]
 [   426    0]]
```

Graph Visualization

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.figure(figsize=(3,3))
sns.countplot(x='is_fraud', data=data)
plt.title("Fraud vs Genuine Transactions")
plt.xlabel("Transaction Type")
plt.ylabel("Count")
plt.xticks([0,1], ['Genuine', 'Fraud'])
plt.show()
```

